THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 84	Use Heron's formula.
2	**(a)** 21	$s = \frac{a+b+c}{2}$.
3	**(a)** 616	Find r from $C = 2\pi r$, then use $A = \pi r^2$.
4	**(a)** 176	Distance in one turn = circumference.
5	**(d)** 70	Area $= \frac{1}{2}d_1d_2$.
6	**(c)** 36	Use Brahmagupta's formula $\sqrt{(s-a)(s-b)(s-c)(s-d)}$.
7	**(b)** 44	Arc $= \frac{\theta}{360} \times 2\pi r$.
8	**(b)** 22	Perimeters are equal.
9	**(a)** 42	Use Heron's formula.
10	**(c)** 18	$s = \frac{a+b+c}{2}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$s = 21$ cm (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 84$ cm^2 (1).	1+1
12	Circumference $= 176$ cm (1); Revolutions $= \frac{4400}{176} = 25$ (1).	1+1
13	Arc $= \frac{120}{360} \times 2\pi r$ (1); $= 44$ cm (1).	1+1
14	$s = 21$ cm (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 42$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$s = 21$ cm (1); $A = 84$ cm^2 (1); $h = \frac{2A}{15} = \frac{168}{15} = \frac{56}{5}$ cm (1).	1+1+1
16	$s = 13$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ (1); $A = 36$ cm^2 (1).	1+1+1
17	Parallelogram: $A = 14 \times 9 = 126$ cm^2 (1); Triangle: $\frac{1}{2} \times \text{base} \times 18 = 126$ (1); base $= 14$ cm (1).	1+1+1
	OR: $s = 24$ cm (1); $A = 84$ cm^2 (1); $h = \frac{2A}{21} = \frac{168}{21} = 8$ cm (1).	1+1+1
18	$s = 24$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ (1); $A = 80$ cm^2 (1).	1+1+1
	OR: Parallelogram: $A = 18 \times 7 = 126$ cm^2 (1); Triangle: $\frac{1}{2} \times \text{base} \times 14 = 126$ (1); base $= 18$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Outer area $= \frac{22}{7} \times 21^2 = 1386$ m^2 (1); Inner area $= 616$ m^2 (1); Track area $= 770$ m^2 (1); Cost $= 770 \times 5$ (1); $=$ ₹3850 (1).	1+1+1+1+1
	OR: Outer length $= 34$ m, outer breadth $= 24$ m (1); Outer area $= 816$ m^2 (1); Lawn area $= 600$ m^2 (1); Path area $= 216$ m^2 (1); Units stated (1).	1+1+1+1+1
20	Outer area $= \frac{22}{7} \times 28^2 = 2464$ m^2 (1); Inner area $= 1386$ m^2 (1); Track area $= 1078$ m^2 (1); Cost $= 1078 \times 4$ (1); $=$ ₹4312 (1).	1+1+1+1+1
	OR: Outer length $= 46$ m, outer breadth $= 31$ m (1); Outer area $= 1426$ m^2 (1); Lawn area $= 1000$ m^2 (1); Path area $= 426$ m^2 (1); Units stated (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up circumference ($2\pi r$) and area (πr^2), or writing the area in cm instead of cm^2.
- In Heron's formula, using the perimeter instead of the semi-perimeter s in the factors $(s-a)$.
- Using a slanting side instead of the perpendicular height in the area of a parallelogram or triangle.
- Treating the width of a path as the outer radius; the outer radius is $r + w$ and the area of the path is the *difference* of the two circle areas.
- Forgetting the factor $\frac{\theta}{360}$ in the arc length formula, or using the area formula πr^2 for arc length.
- Applying Brahmagupta's formula to a quadrilateral that is not cyclic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1386	Find r from $C = 2\pi r$, then use $A = \pi r^2$.
2	**(d)** 220	Distance in one turn = circumference.
3	**(b)** 96	Area $= \frac{1}{2}d_1d_2$.
4	**(c)** 80	Use Brahmagupta's formula $\sqrt{(s-a)(s-b)(s-c)(s-d)}$.
5	**(d)** 22	Arc $= \frac{\theta}{360} \times 2\pi r$.
6	**(a)** 44	Perimeters are equal.
7	**(d)** 36	Use Heron's formula.
8	**(d)** 16	$s = \frac{a+b+c}{2}$.
9	**(a)** 3850	Find r from $C = 2\pi r$, then use $A = \pi r^2$.
10	**(a)** 264	Distance in one turn = circumference.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Circumference $= 220$ cm (1); Revolutions $= \frac{8800}{220} = 40$ (1).	1+1
12	Arc $= \frac{90}{360} \times 2\pi r$ (1); $= 22$ cm (1).	1+1
13	$s = 18$ cm (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 36$ cm^2 (1).	1+1
14	Circumference $= 132$ cm (1); Revolutions $= \frac{6600}{132} = 50$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$s = 18$ cm (1); $A = 36$ cm^2 (1); $h = \frac{2A}{17} = \frac{72}{17} = \frac{72}{17}$ cm (1).	1+1+1
16	$s = 26$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ (1); $A = 144$ cm^2 (1).	1+1+1
17	Parallelogram: $A = 22 \times 5 = 110$ cm^2 (1); Triangle: $\frac{1}{2} \times \text{base} \times 10 = 110$ (1); base $= 22$ cm (1).	1+1+1
	OR: $s = 21$ cm (1); $A = 42$ cm^2 (1); $h = \frac{2A}{20} = \frac{84}{20} = \frac{21}{5}$ cm (1).	1+1+1
18	$s = 29$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ (1); $A = 180$ cm^2 (1).	1+1+1
	OR: Parallelogram: $A = 16 \times 12 = 192$ cm^2 (1); Triangle: $\frac{1}{2} \times \text{base} \times 24 = 192$ (1); base $= 16$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Outer area $= \frac{22}{7} \times 42^2 = 5544$ m^2 (1); Inner area $= 2464$ m^2 (1); Track area $= 3080$ m^2 (1); Cost $= 3080 \times 3$ (1); $=$ ₹9240 (1).	1+1+1+1+1
	OR: Outer length $= 54$ m, outer breadth $= 34$ m (1); Outer area $= 1836$ m^2 (1); Lawn area $= 1500$ m^2 (1); Path area $= 336$ m^2 (1); Units stated (1).	1+1+1+1+1
20	Outer area $= \frac{22}{7} \times 42^2 = 5544$ m^2 (1); Inner area $= 3850$ m^2 (1); Track area $= 1694$ m^2 (1); Cost $= 1694 \times 6$ (1); $=$ ₹10164 (1).	1+1+1+1+1
	OR: Outer length $= 44$ m, outer breadth $= 32$ m (1); Outer area $= 1408$ m^2 (1); Lawn area $= 864$ m^2 (1); Path area $= 544$ m^2 (1); Units stated (1).	1+1+1+1+1

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- Using a slanting side instead of the perpendicular height in the area of a parallelogram or triangle.
- Treating the width of a path as the outer radius; the outer radius is $r + w$ and the area of the path is the *difference* of the two circle areas.
- Forgetting the factor $\frac{\theta}{360}$ in the arc length formula, or using the area formula πr^2 for arc length.
- Applying Brahmagupta's formula to a quadrilateral that is not cyclic.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 135	Area $= \frac{1}{2}d_1d_2$.
2	**(d)** 144	Use Brahmagupta's formula $\sqrt{(s-a)(s-b)(s-c)(s-d)}$.
3	**(d)** 110	Arc $= \frac{\theta}{360} \times 2\pi r$.
4	**(b)** 66	Perimeters are equal.
5	**(d)** 84	Use Heron's formula.
6	**(a)** 33	$s = \frac{a+b+c}{2}$.
7	**(b)** 5544	Find r from $C = 2\pi r$, then use $A = \pi r^2$.
8	**(b)** 352	Distance in one turn = circumference.
9	**(c)** 240	Area $= \frac{1}{2}d_1d_2$.
10	**(b)** 180	Use Brahmagupta's formula $\sqrt{(s-a)(s-b)(s-c)(s-d)}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Arc $= \frac{180}{360} \times 2\pi r$ (1); $= 110$ cm (1).	1+1
12	$s = 24$ cm (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 84$ cm^2 (1).	1+1
13	Circumference $= 88$ cm (1); Revolutions $= \frac{8800}{88} = 100$ (1).	1+1
14	Arc $= \frac{45}{360} \times 2\pi r$ (1); $= 22$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$s = 16$ cm (1); $A = 24$ cm^2 (1); $h = \frac{2A}{15} = \frac{48}{15} = \frac{16}{5}$ cm (1).	1+1+1
16	$s = 30$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ (1); $A = 210$ cm^2 (1).	1+1+1
17	Parallelogram: $A = 25 \times 6 = 150$ cm^2 (1); Triangle: $\frac{1}{2} \times \text{base} \times 12 = 150$ (1); base $= 25$ cm (1).	1+1+1
	OR: $s = 16$ cm (1); $A = 48$ cm^2 (1); $h = \frac{2A}{12} = \frac{96}{12} = 8$ cm (1).	1+1+1
18	$s = 32$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ (1); $A = 180$ cm^2 (1).	1+1+1
	OR: Parallelogram: $A = 20 \times 8 = 160$ cm^2 (1); Triangle: $\frac{1}{2} \times \text{base} \times 16 = 160$ (1); base $= 20$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Outer area $= \frac{22}{7} \times 28^2 = 2464$ m^2 (1); Inner area $= 616$ m^2 (1); Track area $= 1848$ m^2 (1); Cost $= 1848 \times 2$ (1); $=$ ₹3696 (1).	1+1+1+1+1
	OR: Outer length $= 51$ m, outer breadth $= 41$ m (1); Outer area $= 2091$ m^2 (1); Lawn area $= 1575$ m^2 (1); Path area $= 516$ m^2 (1); Units stated (1).	1+1+1+1+1
20	Outer area $= \frac{22}{7} \times 49^2 = 7546$ m^2 (1); Inner area $= 5544$ m^2 (1); Track area $= 2002$ m^2 (1); Cost $= 2002 \times 5$ (1); $=$ ₹10010 (1).	1+1+1+1+1
	OR: Outer length $= 70$ m, outer breadth $= 50$ m (1); Outer area $= 3500$ m^2 (1); Lawn area $= 2400$ m^2 (1); Path area $= 1100$ m^2 (1); Units stated (1).	1+1+1+1+1

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